

Bangladesh University of Business & Technology (BUBT)



LAB Report

Course Code : CSE 207
Course Title : Database Systems Lab
Experiment No. : 01

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ID: 20245103035
Intake: 53
Section: 01
Program: B.Sc. Engg. in CSE

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Date of Submission: 13/08/2025

Signature of Teacher

1. Create a DB named "BUBT_DB1".

```
CREATE DATABASE `fahim_035`;
```

Show query box

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0011 seconds.)

```
CREATE DATABASE `fahim_035`;
```

[Edit inline] [Edit] [Create PHP code]

⚠ Error: #1046 No database selected

2. Create three tables in this DB named "Student", "Faculty", "Course".

```
CREATE TABLE `Student`(`Name` varchar(20), `St_Id` int(3), `NID` int(4), `Intake` int(3), `Section` int(2), `CGPA` float(3,2));
```

```
CREATE TABLE `Faculty`(`Name` varchar(20), `Short code` varchar(4), `Course Code` varchar(8), `Conduct Semester` varchar(10));
```

```
CREATE TABLE `Course`(`Course Title` varchar(20), `Course Code` varchar(8));
```

Name	St_ID	NID	Intake	section	CGPA
Ashraful	1	102	44	1	3.3
Tonny	2	101	44	1	3.5
Zakir	33	103	35	2	3.98
Urmi	6	104	35	1	3.96
Tonny	23	105	44	1	3.5

Name	Short_code	Course_code	Conduct_semester
Farha Akhter Munmun	FAM	CSE 319	Summer 2019
Sweety Lima	SWL	CSE 209	Fall 2019-20
Shumi Khatun	SKN	NULL	Summer 2019
Nadia Afrin Ritu	NULL	CSE 351	NULL

Course_title	Course_code
Database Systems	CSE 207
Computer Networks	CSE 310
Data Communication	CSE 209
System Analysis Design	NULL

3(a). Update the short code whose short code is NULL.

```
UPDATE `faculty` SET `Short code`='NAR' WHERE `Name`='Nadia Afrin Ritu';
```

Name	Short_code	Course_code	Conduct_semester
Farha Akhter Munmun	FAM	CSE 319	Summer 2019
Nadia Afrin Ritu	NAR	CSE 351	NULL
Shumi Khatun	SKN	NULL	Summer 2019
Sweetly Lima	SWL	CSE 209	Fall 2019-20

3(b).Update the course code that is NULL.

```
UPDATE `faculty` SET `Course Code`='CSE 207' WHERE `Short code`='SKN';  
UPDATE `course` SET `Course Code`='CSE 351' WHERE `Course Title`='System  
Analysis Design';
```

Course_title	Course_code
Database Systems	CSE 207
Data Communication	CSE 209
Computer Networks	CSE 310
System Analysis Design	CSE351

Name	Short_code	Course_code	Conduct_semester
Farha Akhter Munmun	FAM	CSE 319	Summer 2019
Nadia Afrin Ritu	NAR	CSE 351	NULL
Shumi Khatun	SKN	CSE 207	Summer 2019
Sweetly Lima	SWL	CSE 209	Fall 2019-20

3(c). Update the conduct semester that is NULL.

```
UPDATE `faculty` SET `Conduct Semester`='Summer 2020' WHERE `Short  
code`='NAR';
```

Name	Short_code	Course_code	Conduct_semester
Farha Akhter Munmun	FAM	CSE 319	Summer 2019
Nadia Afrin Ritu	NAR	CSE 351	Summer 2020
Shumi Khatun	SKN	CSE 207	Summer 2019
Sweetly Lima	SWL	CSE 209	Fall 2019-20

3(d). UAdd a new column named “Age” in student relation.

```
ALTER TABLE student ADD COLUMN `Age` int(2);
```

Name	St_ID	NID	Intake	section	CGPA	Age
Ashraful	1	102	44	1	3.3	NULL
Tonny	2	101	44	1	3.5	NULL
Urmi	6	104	35	1	3.96	NULL
Tonny	23	105	44	1	3.5	NULL
Zakir	33	103	35	2	3.98	NULL

3(e). Update the age of each student.

```
UPDATE `student` SET `Age`='22' WHERE `St_Id`='1';
UPDATE `student` SET `Age`='22' WHERE `St_Id`='2';
UPDATE `student` SET `Age`='24' WHERE `St_Id`='33';
UPDATE `student` SET `Age`='24' WHERE `St_Id`='6';
UPDATE `student` SET `Age`='22' WHERE `St_Id`='23';
```

Name	St_ID	NID	Intake	section	CGPA	Age
Ashraful	1	102	44	1	3.3	22
Tonny	2	101	44	1	3.5	22
Urmi	6	104	35	1	3.96	24
Tonny	23	105	44	1	3.5	22
Zakir	33	103	35	2	3.98	24

3(f). Find the names of all Courses in the Course relation.

```
SELECT `Course Title` FROM `course`;
```

Showing rows 0 - 3 (4 total, Query took 0.0002 seconds.)

```
SELECT `Course Title` FROM `course`;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25

Extra options

Course Title

- Database Systems
- Computer Networks
- Data Communication
- System Analysis Design

3(g). Find the names of all teachers who has conducted in summer 2019.

```
SELECT `Name` FROM `faculty` WHERE `Conduct Semester`='Summer 2019';
```

Showing rows 0 - 1 (2 total, Query took 0.0002 seconds.)

```
SELECT `Name` FROM `faculty` WHERE `Conduct Semester`='Summer 2019';
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25

Extra options

Name
Farha Akhter Munmun
Shumi Khatun

☐ Show all | Number of rows: 25

3(h). Find the names of all students whose CGPA is greater than 3.50.

```
SELECT `Name` FROM `student` WHERE `CGPA` > 3.5;
```

Showing rows 0 - 1 (2 total, Query took 0.0001 seconds.)

```
SELECT `Name` FROM `student` WHERE `CGPA` > 3.5;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25

Extra options

Name
Zakir
Urmil

☐ Show all | Number of rows: 25

3(i). Add a new column "Offered semester" after the column "Course Title".

```
ALTER TABLE course ADD COLUMN Offered_Semester varchar (20) AFTER Course_title;
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0003 seconds.)

```
ALTER TABLE course ADD COLUMN Offered_Semester varchar (20) AFTER Course_title;
```

[Edit inline] [Edit] [Create PHP code]

3(j). Find the name and CGPA of each student by their CGPA (Ascending order.).

```
SELECT Name,CGPA FROM student ORDER BY CGPA ASC;
```

Showing rows 0 - 4 (5 total, Query took 0.0001 seconds.) [CGPA: 3.3... - 3.98...]

```
SELECT Name,CGPA FROM student ORDER BY CGPA ASC;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25

Extra options

		Name	CGPA
☐	Edit Copy Delete	Ashraful	3.3
☐	Edit Copy Delete	Tonny	3.5
☐	Edit Copy Delete	Tonny	3.5
☐	Edit Copy Delete	Urmi	3.96
☐	Edit Copy Delete	Zakir	3.98

3(k). Find the name of all students whose age is greater than 22 and CGPA is greater than 3.50.

SELECT Name FROM student WHERE Age > 22 AND CGPA > 3.50;

Showing rows 0 - 1 (2 total, Query took 0.0003 seconds.)

```
SELECT Name FROM student WHERE Age > 22 AND CGPA > 3.50;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25

Extra options

		Name
☐	Edit Copy Delete	Urmi
☐	Edit Copy Delete	Zakir

3(l). Find the Id of all those students who are from 44 intake or CGPA is less than 3.40.

SELECT Name FROM student WHERE Intake = 44 OR CGPA <3.40;

Showing rows 0 - 2 (3 total, Query took 0.0003 seconds.)

```
SELECT Name FROM student WHERE Intake = 44 OR CGPA <3.40;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25

Extra options

		Name
☐	Edit Copy Delete	Ashraful
☐	Edit Copy Delete	Tonny
☐	Edit Copy Delete	Tonny

3(m). Find the Id of all those students whose CGPA is 3.30 or 3.50 or 3.96.

SELECT St_Id FROM Student WHERE CGPA IN (3.30, 3.50, 3.96);

✓ Showing rows 0 - 1 (2 total, Query took 0.0001 seconds.)

```
SELECT St_Id FROM Student WHERE CGPA IN (3.30, 3.50, 3.96);
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 ▼

Extra options

				St_Id
☐	 Edit	 Copy	 Delete	2
☐	 Edit	 Copy	 Delete	23